BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, GUJRANWALA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

COMPUTER SCIENCE

Past Paper 2024 • Paper Code: 8837 / 8836

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Computer Science (Intermediate Part-II) Paper : II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 8837

Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

QMS-ACCESS. MS-ACCESS

Q1. 1. An entity related to itself in an ERD model refers to relationship:

(A) recursive
(B) one to many
(C) many to many
(D) one to one

Q2. The graphical query tool is known as:

(A) query grid
(B) design grid
(C) query form
(D) design form

Q3. A virtual table that is constructed from other tables is called:

(A) table
(B) tuple
(C) view
(D) relation

Q4. The data that causes inconsistency lacks:

(A) good data
(B) data integrity
(C) data redundancy
(D) data anomaly

Q5. Referential integrity is applied on:

(A) primary key
(B) composite key
(C) sort key
(D) foreign key

Q6. Each separate piece of information stored in a record is called:

(A) form
(B) field
(C) table
(D) query

QC-LANGUAGE. C-LANGUAGE

Q7 (C). If x = 2 , y = 3 , the expression (x > y)? (x + y) : (x * y); will display output as:

(A) 5
(B) 10
(C) 6
(D) 1

Q8 (C). The keyword "break" cannot be used within:

(A) do while
(B) if – else
(C) for
(D) while

Q9 (C). The programmer usually prepares the source code using

(A) compiler
(B) text editor
(C) debugger
(D) linker

Q10 (C). Memory is allocated to a local variable at the time of its:

(A) declaration
(B) destruction
(C) definition
(D) first reference

Q11 (C). The character conversion may occur in:

(A) text stream
(B) binary stream
(C) input stream
(D) output stream

Q12 (C). Escape Sequence used to insert a Tab in C-Language is:

(A) \a
(B) \b
(C) \t
(D) \n

Q13 (C). Format specifier %5.3 for the value 4.1257, will produce the output:

(A) 4.125
(B) 4.126
(C) □4.126
(D) □4.125

Q14 (C). Conditional operator is an alternative of:

(A) if
(B) if – else
(C) switch
(D) for

Q15 (C). The data type in C-language that can handle fractional values, is called:

(A) long
(B) char
(C) float
(D) int

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q7 (VB). Data can be entered in a control when it has:

(A) tab order
(B) mask
(C) focus
(D) control

Q8 (VB). In Visual Basic, a form is another name for:

(A) object
(B) window
(C) property
(D) label

Q9 (VB). Which of the following is not a control?

(A) textbook
(B) close button
(C) command button
(D) option button

Q10 (VB). The property that changes text color is:

(A) backcolor
(B) textcolor
(C) forecolor
(D) fontcolor

Q11 (VB). Visual Basic is a product of:

(A) Microsoft
(B) IBM
(C) Hewlet
(D) Sun System

Q12 (VB). A control that provides an identifiable grouping for control is:

(A) textbox
(B) button
(C) label
(D) listbox

Q13 (VB). The output of compiler is:

(A) source code
(B) object code
(C) linked code
(D) program

Q14 (VB). A predefined screen that can be used as one of the building blocks of an application:

(A) object
(B) window
(C) property
(D) label

Q15 (VB). The code that is used in a sub procedure to stop execution of a program is:

(A) exit
(B) halt
(C) end
(D) end sub

Computer Science (Intermediate Part-II) Paper : II (Subjective Type)

Time Allowed: 2.10 hours | **Maximum Marks:** 60

> Note: Section I is compulsory.

QMS-ACCESS. MS-ACCESS

Q2. (Marks 12) Write short answers to any SIX questions:

i. Write down two advantages of database systems.

ii. In which condition, you need to create composite key in a relation? Justify your answer.

iii. What is foreign key?

iv. Differentiate between relation and relationship.

v. What is the main purpose of "Partitioned" data distribution strategy?

vi. What is meant by "atomic values"?

vii. What is database wizard?

viii. Why are we using Input Masks?

ix. Differentiate between ListBox and ComboBox.

QC-LANGUAGE. C-LANGUAGE

Q3 (C-LANGUAGE). (Marks 12) Write short answers to any SIX questions:

i. Define variable initialization.

ii. Write down the following statement in C-language. "Age is between 18 years and 50 years"

iii. Find any two errors: integer x , y, z = x + y; printf("z = %d" , z);

iv. Find output: int a = 5 , b = 10; int z = a + b * 2; printf("Result = %d" , z);

v. What is meant by & in scanf function?

vi. Find any two errors: void main (); {int a = 10; printf("%s" , b); }

vii. Write the output of the following code: printf("\n Amir \t\t 78 \t\t 425"); printf("\n Tahir \t\t 23 \t\t 388");

viii. Describe the purpose of "\0" in strings.

ix. What is a stream?

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q3 (VISUAL BASIC). (Marks 12) Write short answers to any SIX questions:

i. Define a method in VB.

ii. What is the main advantage of using Visual Basic?

iii. Describe form window.

iv. Differentiate between menubar and toolbar.

v. Write down any two forms events.

vi. How is a picture removed at runtime?

vii. What is ListIndex property?

viii. Define WordWrap in VB.

ix. What is FileListBox control?

QC-LANGUAGE. C-LANGUAGE

Q4 (C-LANGUAGE). (Marks 12) Write short answers to any SIX questions:

i. Why break statement is used in a switch ( ) structure?

ii. Predict the output of the following code segment: int a = 10 , b = 5; if (a > b) printf("PUNJAB"); else printf("KPK");

iii. Define nested loop. How does it work?

iv. Convert the following code into while loop: int i; for (i = 1 ; i <=5 ; i++) { printf("In Pakistan"); }

v. Predict the output of the following code segment: int n; for (n = 10; n>=1 ; n––) printf("%d",n);

vi. Define function.

vii. How is a function activated?

viii. Write down the use of formal parameters.

ix. Write down two importances of functions?

> Note: (Turn Over)

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q4 (VISUAL BASIC). (Marks 12) Write short answers to any SIX questions:

i. Define currency data type in VB.

ii. Describe option explicit declaration in VB.

iii. Write down the syntax of input box.

iv. How select.....Case help in coding?

v. What is default property of command button?

vi. Trace the output: if(10 MOD 3 = 0) Then print ("YES") else print ("NO") End if

vii. How an AND operator is used in VB statement?

viii. What is sentinal value?

ix. Write down the syntax of For.....Next loop.

> Note: SECTION - II

QMS-ACCESS. MS-ACCESS

> Note: Note: Attempt any ONE question.

Q5.

[8] Briefly describe four major components of database systems.

Q6.

[8] Explain different types of Queries?

> Note: SECTION – III

> Note: Note: Attempt any TWO questions (either from "C-Language" or from "Visual Basic"):

QC – LANGUAGE. C – LANGUAGE

Q7 (C).

[8] What are errors? Discuss different types of errors which may occur in C-Language program.

Q8 (C).

[8] Write a program in C-language that inputs a year and finds whether it is a leap year or not, using if-else structure.

Q9 (C).

[8] Write a program that displays first five odd numbers and their average using while loop.

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q7 (VB).

[8] Discuss any eight form objects in VB.

Q8 (VB).

[8] Write a program in Visual Basic that inputs a year and finds whether it is a leap year or not, using If...Then...Else structure.

Q9 (VB).

[8] Write a program in VB that displays first five odd numbers and their average using While Wend loop.

GROUP – II

Computer Science (Intermediate Part-II) Paper : II (Objective Type)

Time Allowed: 20 Minutes	**Maximum Marks:** 15	**Paper Code:** 8836

> Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

QMS-ACCESS. MS-ACCESS

Q1. 1. The smallest meaningful unit of data in a database is called:

(A) field
(B) character
(C) record
(D) byte

Q2. In relational database, a table is also called:

(A) tuple
(B) relation
(C) schema
(D) file

Q3. Which of the following data model is more flexible?

(A) Network data model
(B) Relational data model
(C) Hierarchical data model
(D) Basic data model

Q4. Default field size of text data type field in MS Access:

(A) 20
(B) 30
(C) 40
(D) 50

Q5. In an ER diagram, a rectangle represents:

(A) entity
(B) relationship
(C) attribute
(D) action

Q6. The columns of a table correspond to:

(A) cell
(B) table
(C) record
(D) field

QC-LANGUAGE. C-LANGUAGE

Q7 (C). The case block ends with

(A) End Select
(B) End Case
(C) break
(D) Case else

Q8 (C). Format Specifier %5.2f for value 4.12157, will produce output:

(A) □4.12
(B) □□4.12
(C) □□□4.12
(D) □5.121

Q9 (C). If a = 70 what will be the output of the following expression? n = (a >= 80)? 10 : 20;

(A) 5
(B) 10
(C) 15
(D) 20

Q10 (C). The name of header file is written between:

(A) [ ]
(B) ( )
(C) < >
(D) << >>

Q11 (C). In which loop the condition comes after the body of the loop?

(A) for loop
(B) while loop
(C) do – while loop
(D) Infinite loop

Q12 (C). Single line comment starts with

(A) //
(B) \\
(C) *\
(D) *

Q13 (C). A file is stored in

(A) RAM
(B) Hard Disk
(C) ROM
(D) Cache

Q14 (C). Escape sequence used for backspace in C

(A) \a
(B) \b
(C) \t
(D) \r

Q15 (C). Local variables are also called

(A) automatic variables
(B) normal variables
(C) global variables
(D) basic variables

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q7 (VB). A procedure or function that imparts some action to an object is called

(A) property
(B) control
(C) method
(D) label

Q8 (VB). Methods characterize an object's

(A) behaviour
(B) appearance
(C) value
(D) style

Q9 (VB). Which event detects the pressing of standard displayable characters?

(A) key down
(B) key up
(C) keyboard
(D) key press

Q10 (VB). The diamond in a flowchart indicates

(A) input
(B) output
(C) condition
(D) progress

Q11 (VB). The function which is used to display a message

(A) input box
(B) message
(C) msgbox
(D) text box

Q12 (VB). Control, that is used to display text or to accept user input

(A) label
(B) textbox
(C) button
(D) property

Q13 (VB). Relational operators are used to

(A) add
(B) compare
(C) multiply
(D) divide

Q14 (VB). Forms and controls are collectively recognized as

(A) properties (B) objects

(C) methods (D) events

Q15 (VB). The default message box style is

(A) OK only (B) Yes No

(C) Yes No Cancel (D) Retry Cancel

Computer Science (Intermediate Part-II) Paper : II (Subjective Type)

Time Allowed: 2.10 hours | **Maximum Marks:** 60

> Note: Section I is compulsory.

QMS-ACCESS. MS-ACCESS

Q2. (Marks 12) Write short answers to any SIX questions:

i. Define the term database.

ii. Discuss the advantage of variable length field over fixed length field.

iii. What is the role of end user in database system?

iv. Briefly describe the concept of cardinality.

v. Define centralized data distribution strategy.

vi. What is synonym problem in database?

vii. Briefly describe the role of table object in database.

viii. Define the concept of wildcards.

ix. Discuss the layout of datasheet form.

QC-LANGUAGE. C-LANGUAGE

Q3 (C-LANGUAGE). (Marks 12) Write short answers to any SIX questions:

i. What is meant by initializing a variable?

ii. Write the following statement in C-language code: Temp between 15 and 32 inclusive

iii. Write errors in the following code: int m, y ; m = 10.3; y = 5; printF("%d %d", m , y).

iv. Predict the output of the following code: int x = 5 , y = 10; x = x + y; y = y – x; printf("%d %d", x , y);

v. Why is ampersand (&) used in scanf function?

vi. Find errors in the following code: #include <stdio.h> viode main () { int a = 10; printf("%d" , a); }

vii. Predict output of the following code: int n = 512 , b ; b = n% 10; printf("%d \n %d", n , b);

viii. Define binary stream.

ix. Write usage of file pointer.

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q3 (VISUAL BASIC). (Marks 12) Write short answers to any SIX questions:

i. What are three basic controls used in Visual Basic?

ii. What is properties windows?

iii. Write ways to exit from Visual Basic.

iv. How object methods are called?

v. Define event procedures.

vi. What is meant by Focus?

vii. Differentiate between Modal and Modeless Forms

viii. What is the importance of visible property in VB Form?

ix. Write the method to close a VB Form.

QC-LANGUAGE. C-LANGUAGE

Q4 (C-LANGUAGE). (Marks 12) Write short answers to any SIX questions:

i. Give examples of two control structures.

ii. Predict the output of the code: char ch; ch = 'p'; if(ch == 'p') printf("pass"); else printf("fail");

iii. Differentiate between 'do – while' and 'while' loop.

iv. Convert the following for( ) loop in to while ( ): for(int i = 1 ; i <= 5 ; i++) printf("%d \n" , i);

v. Give output of the following code: for(int x=1 ; x <= 5 ; x++) printf("%d \n" , x * x);

vi. Give any two advantages of using a user defined function in C-language program.

vii. What is the purpose of function call and function return?

viii. Differentiate between local and global variables.

ix. What is the purpose of function header and function prototype?

> Note: (Turn Over)

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q4 (VISUAL BASIC). (Marks 12) Write short answers to any SIX questions:

i. Differentiate between explicit and implicit declaration of variable.

ii. Write the syntax of input box.

iii. List four relational operators in VB.

iv. What symbol is used for NOT operator in Visual Basic? Give example.

v. When should a for – Next loop be used?

vi. Write syntax of for – Next loop.

vii. What is the purpose of counter variable?

viii. Write output of following code segment: x = 0 while x <= 15 print x + 2 x = x + 2 wend

ix. What is meant by iteration in a loop?

> Note: SECTION - II

QMS-ACCESS. MS-ACCESS

> Note: Note: Attempt any ONE question.

Q5.

[8] Define Database model? Explain Hierarchical model along with example.

Q6.

[8] Define view. What are two table views available in MSAccess? How can you switch between them?

> Note: SECTION – III

> Note: Note: Attempt any TWO questions (either from "C-Language" or from "Visual Basic"):

QC – LANGUAGE. C – LANGUAGE

Q7 (C).

[8] How would you create, edit, compile, link and execute a program in C-Language? Discuss briefly.

Q8 (C).

[8] Write a program in C-language that inputs three numbers and finds the smallest number.

Q9 (C).

[8] Write a program in C-language that calculates the sum of the following series using loop: 1.0 1.5 2.0 2.5 3.0 3.5 ..... 10.0

> Note: OR

QVISUAL BASIC. VISUAL BASIC

Q7 (VB).

[8] How would you create, edit, compile, link and execute a program in VB? Discuss briefly.

Q8 (VB).

[8] Write a program in Visual Basic that inputs three numbers and finds the smallest number.

Q9 (VB).

[8] Write a program in Visual Basic that calculates the sum of the following series using loop: 1.0 1.5 2.0 2.5 3.0 3.5 ..... 10.0